AURIGENE

PHARMACEUTICAL SERVICES

PROJECT REPORT

ON

PEPTIDE AGAINST ANTHOCYANINS

Date : 19th Dec 2025

Presented To:

Fabian Falkenberg

fabian.falkenberg@henkel.com

Presented by:

Asha Telkar

asha.telkar@henkel.com

Research & Informatics Team

TABLE OF CONTENTS

SI.No	Contents	Page Number
1	Title	2
2	Objective	2
3	Methods	3
4	Results & Discussion	3
5	Conclusion	14

Title

Peptide against anthocyanins

Objective

To design the peptides based strong specific chelator for Fe^{3+} in colorants (anthocyanins, polyphenolics).

Methods

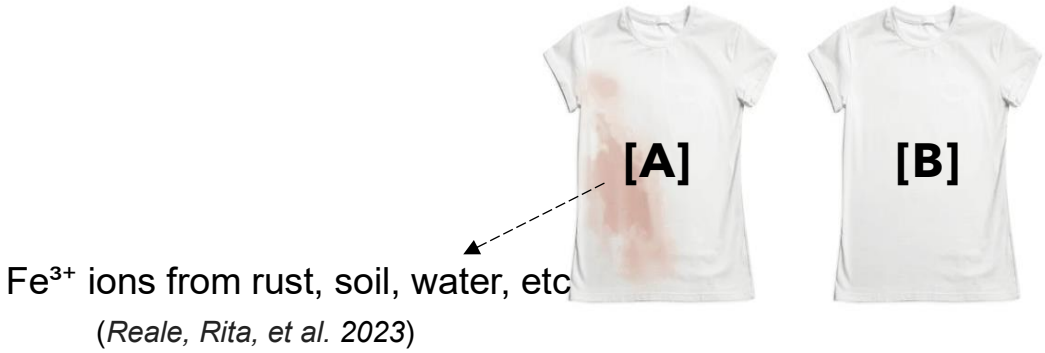
The peptides are developed and validated by using combination of bioinformatics analysis, molecular modelling approaches (such as molecular docking, MD simulations) and AI related techniques such as reinforcement learning.

Results & Discussion

The Core Stain Removal Mechanism

Anthocyanin pigments in plant-based residues exist in a colorless/red flavylium cation form. In the presence of Fe^{3+} ions from environmental sources (water, metal surfaces, or food residues), these pigments undergo coordination chemistry forming stable anthocyanin-Fe^{3+} complexes. This metallo-organic complex exhibits intense blue-black coloration and strong surface adherence through Fe^{3+}-mediated crosslinking to substrates, resulting in persistent staining shown in Fig, 1 [A].

Engineered Fe^{3+}- chelating peptides competitively sequester Fe^{3+} ions through superior binding affinity. This metal ion stripping disrupts the stain complex, liberating soluble anthocyanin that washes away, while forming stable, colorless peptide-Fe^{3+} complexes shown in Fig, 1 [B].

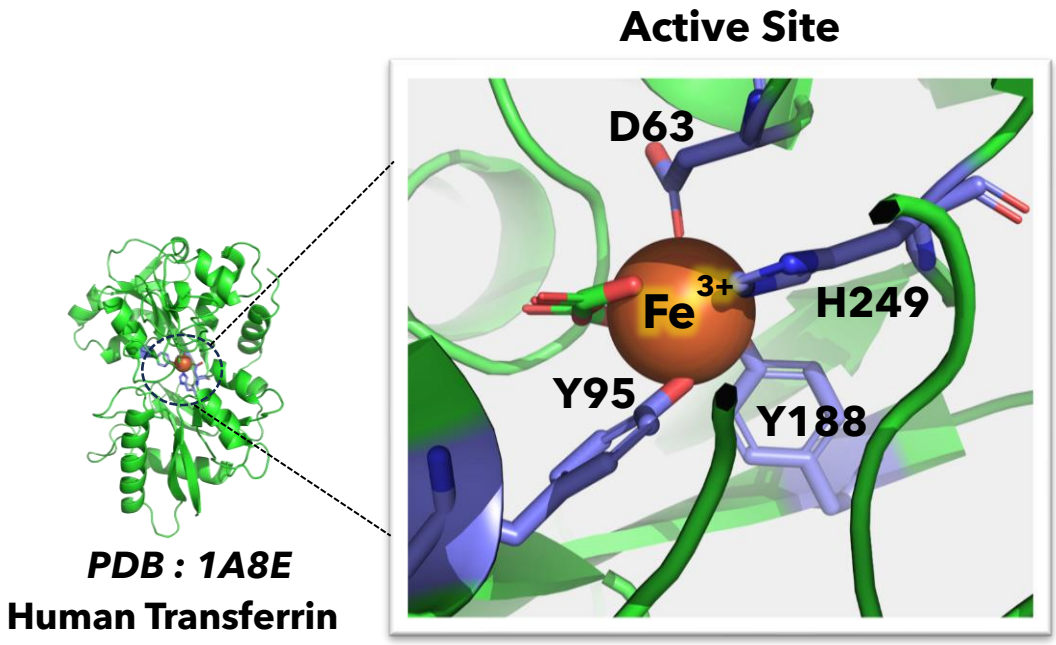


Fig, 1. The image demonstrates a rust-stained garment [A] into a clean garment [B].

Biological Blueprint (Inspired by Nature)

Active site coordination and residue selection for peptide design were inspired by structural analysis of a reference peptide obtained from the Protein Data Bank (PDB : 1A8E). The binding orientation and coordination geometry of Fe^{3+} were carefully examined, particularly its interaction with key active residues specifically D63, H249, Y95, and Y188.

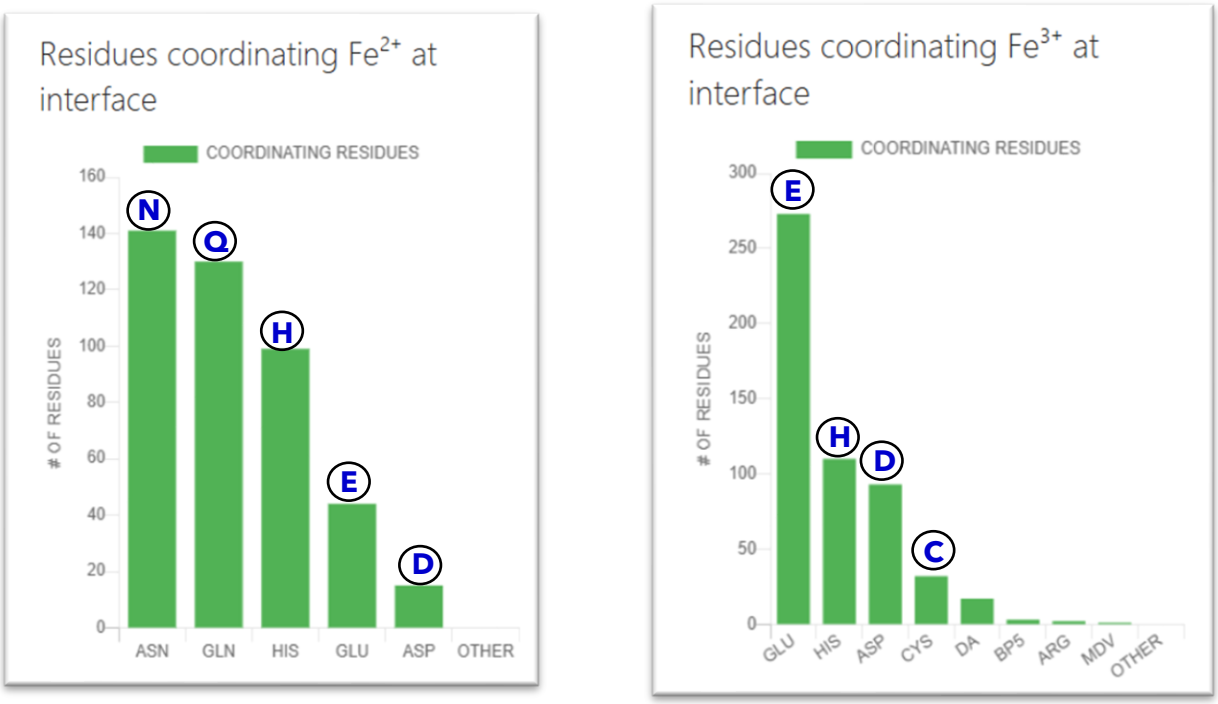
These residues were identified as critical for metal coordination and stabilization. Their spatial arrangement and functional roles in the reference structure were used to guide the rational design and development of the target peptide, ensuring preservation of the essential Fe^{3+} binding environment.



Fig, 2. A detailed view of the Fe^{3+} - binding active site of Human Transferrin (PDB ID: 1A8E), illustrating the native biological mechanism for iron chelation.

Fe^{2+} 's & Fe^{3+} 's coordinated residues

Utilizing the comprehensive structural data of Fe^{3+} -binding sites from InterMetalDB, our peptide design strategy is centered on engineering synthetic motifs that accurately mimic the ion's preferred coordination sphere. These analyses often show the Fe^{3+} ion adopting an octahedral geometry, with the coordination sphere frequently completed by non-peptide ligands like water or bicarbonate. For peptide design, particularly targeting macromolecular interfaces as highlighted by InterMetalDB, we selected specific Fe^{3+} - coordinating residues, incorporating them into a sequence with precise spacing and context.



Fig, 3. Amino acid proportion of Fe^{2+} 's & Fe^{3+} 's coordinated residues in biomolecules from the PDB database.

Previously reported Fe^{3+} and Fe^{2+} peptides

The design of iron-binding peptides, crucial for developing effective supplements and fortified foods, draws upon sequences derived from diverse natural sources, with distinct motifs preferred for ferric Fe^{2+} and ferrous Fe^{3+} iron.

Fe^{3+} Based Peptides

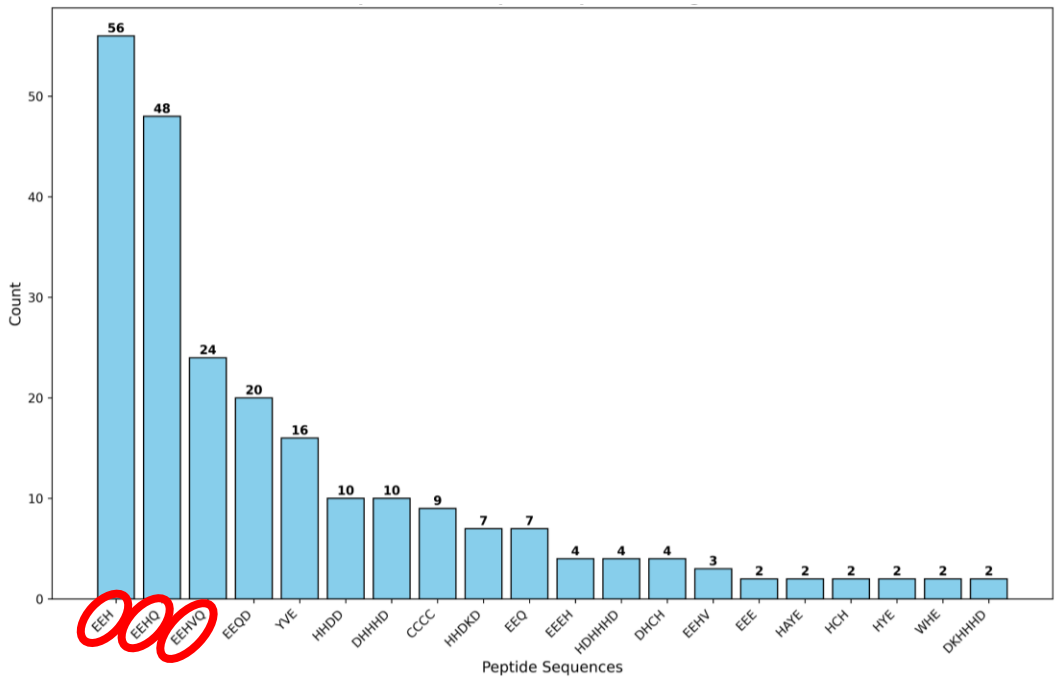
Peptides designed to bind the hard Lewis acid Fe^{3+} are primarily characterized by acidic, oxygen-rich motifs that promote high-affinity chelation and enhance iron solubility. Key sequences, such as the EEEAEAEAR motif from pea protein, are engineered to enhance solubility and bioavailability, while sequences like the acidic EExxED from the CyaY protein family are linked to metal specificity and structural transitions relevant to iron-related diseases. Sequences derived from food proteins, including the 11 kDa (LMNLAIRCRLGPMIGCDLSSDD) and 17 kDa (DNQSEQLEGKEKK) peptides from soybean protein, aim to resolve the "soybean iron paradox" by enhancing absorption. Similarly, whey protein-derived peptides, such as VRTPEVDDE and the extended VEELKPTPEGDLE, serve as effective iron binders for functional food products and supplements, often leveraging multiple carboxylate groups for strong coordination.

Fe^{2+} Based Peptides

In contrast, peptides binding the softer Lewis acid Fe^{2+} often incorporate motifs that utilize a combination of acidic, nitrogen, and flexible residues to promote uptake and bioavailability. Peptides sourced from *Acetes japonicus* DSVNFPVHGL, FKVGQENTPLIK are evaluated as natural supplements with enhanced bioavailability. Peptides from duck egg white, including PVEE, LYAEE, and LHP, focus on improving iron uptake by the body. The effectiveness of Fe^{2+} chelators is significantly correlated with specific amino acid content, as seen in hairtail hydrolysate, where Glutamic acid (Glu) provides primary binding sites, Histidine (His) offers strong coordination, and Glycine (Gly) and Leucine (Leu) contribute to flexible backbone structure and hydrophobic interactions, respectively, all of which are essential for treating iron deficiency anemia.

The most occurred motifs in the PDB database

Based on the analysis of Fe^{3+} coordination at biomolecular interfaces within the PDB, our peptide design focuses on utilizing the most frequently observed continuous binding motifs. The coordination data clearly indicates that Glutamate (E), followed by Histidine (H) and Aspartate (D), are the dominant residues stabilizing Fe^{3+} interfaces. This finding is directly reflected in the most common peptide sequences in a globular orientation identified are EEH, EEHQ, and EEQD. The EE (Glutamate-Glutamate) sequence provides the consistent, high-affinity, dicarboxylate anchor for the hard Fe^{3+} ion, while the inclusion of Histidine (H) or Aspartate (D) in the subsequent position modulates the final coordination geometry, thus optimizing the peptide's ability to chelate the metal and drive desired structural transitions or macromolecular interactions in our final design.

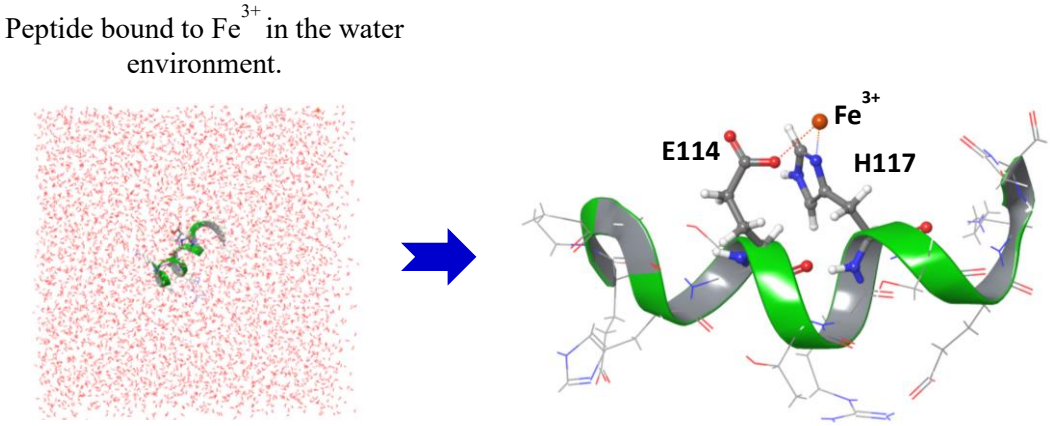


Fig, 4. The most frequent motifs coordinated to Fe^{3+} 's at the interface of biomolecules (PDB Database).

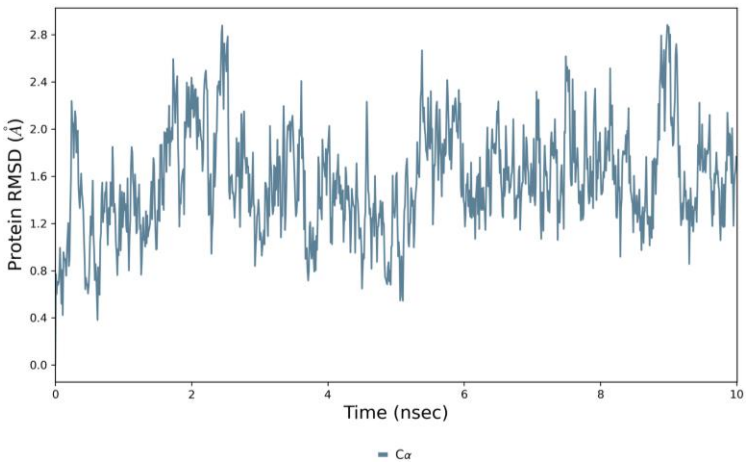
We analyzed the maximum number of PDB files of biomolecules containing Fe^{3+} to understand the interfacial residues interacting with Fe^{3+}.

MD simulation result of generated peptide

Based on this evidence, one of the reference peptides sequence (EPGHSETRHSEK) was generated and subsequently validated using Molecular Dynamics (MD) simulation over 10 ns illustrated in Fig 5. The simulation confirmed that the designed peptide forms a stable alpha-helical structure, effectively coordinating the Fe^{3+}–ion via key residues like E114 and H117 and displacing surrounding water molecules to form a robust chelation complex. The stable Peptide- Fe^{3+} complex, evidenced by a relatively consistent Root Mean Square Deviation (RMSD) plot shown in Fig 6, successfully demonstrates the design strategy's efficacy in creating a high-affinity metal-peptide interaction based on naturally occurring motifs.



Fig, 5. Molecular Dynamics (MD) Simulation Setup of the peptide – Fe^{3+} Complex in an Aqueous Environment (Right), Highlighting the Interaction Between Peptide Residues E114 and H117 with peptide – Fe^{3+} (Left).



Fig, 6. RMSD plot of peptide – Fe^{3+} for 10 ns MD simulation.

Other commonly found motifs reinforcing this trend include ExxH, ExxE, HxxD, and HxD, alongside the EExxED motif identified in the CyaY Protein Family, all emphasizing the critical role of these acidic and nitrogen-containing residues in achieving high-affinity Fe^{3+} binding.

Rules to design the peptides

The design of the Fe^{3+} -binding peptide was guided by a set of strict rules to optimize both chelation specificity and overall peptide stability.

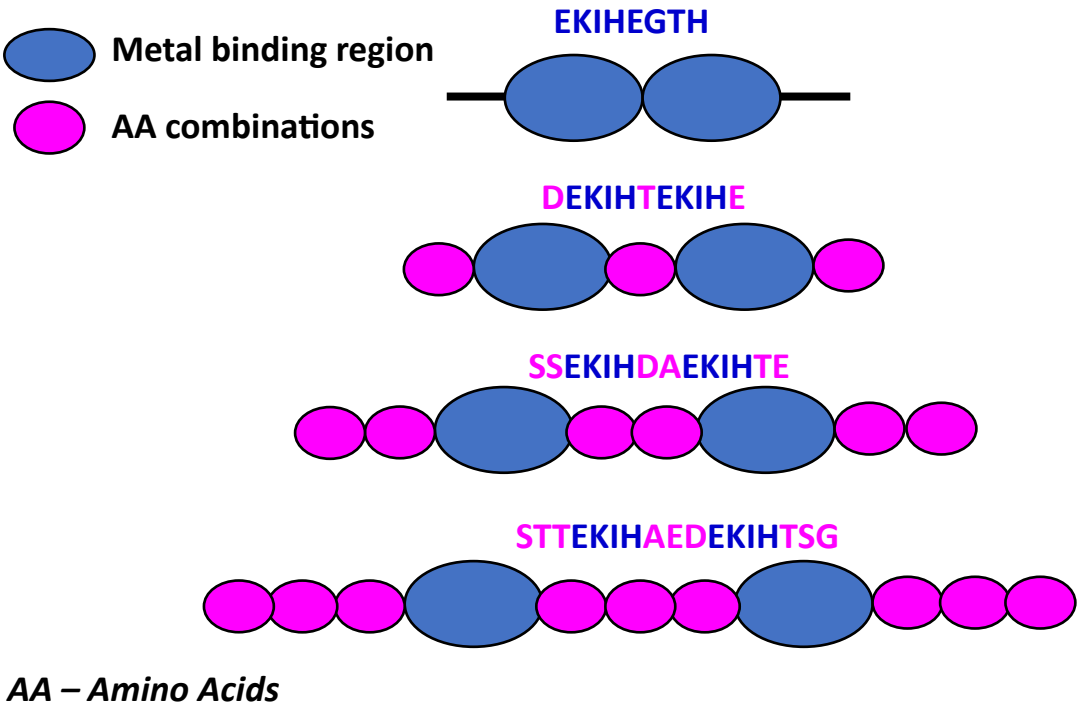
- Peptide should not contain "DG" continuously.
- Should be having hydrophilic amino acids in the spacers/gaps (x)
- The gaps/spacers between the motifs "ExxH", "ExxE", "ExxD", "EExxDD" where xx represents following dipeptides, KI, TG, TA, RE, TY and there can be many combinations of these.
- Replace the X with hydrophilic amino acids and don't include the amino acids N, Q as they are interacting the most with Fe^{2+} we would like to have the peptides to chelate Fe^{3+}
- Peptide should not start with "E" in the N-terminal domain.

Note : Since there is no established reference peptide for liquid-detergent applications, we used biomolecules containing Fe^{3+} as a reference. We confirm that the pKa values of the peptide residues Histidine (~5.5) and Glutamic acid (~4) match well with those of the corresponding residues (His, Glu) in the biomolecules.

Different template peptide designs

Template design 1

The design shown in Fig 7, starts with a core Metal Binding Region, exemplified by the two tetrapeptide motifs sequence EKIHEGTH, which contains the necessary coordinating residues E and H derived from structural analysis. Subsequent designs involve modular expansion of this core structure. Peptides are elongated by flanking the Metal Binding Region with AA combination segments (magenta ovals) and incorporating additional, sometimes internal, metal-binding cores (blue ovals). For instance, the second structure shows a dimerization or tandem arrangement, D E K I H T E K I H E, where AA segments are added to both termini and inserted between two core binding regions. This modular approach allows for the systematic creation of longer, potentially multimeric, peptides S S E K I H D A E K I H T E and S T T E K I H A E D E K I H T S G, enabling the optimization of complex properties like stability, solubility, and the creation of multiple or larger Fe^{3+} coordination sites.

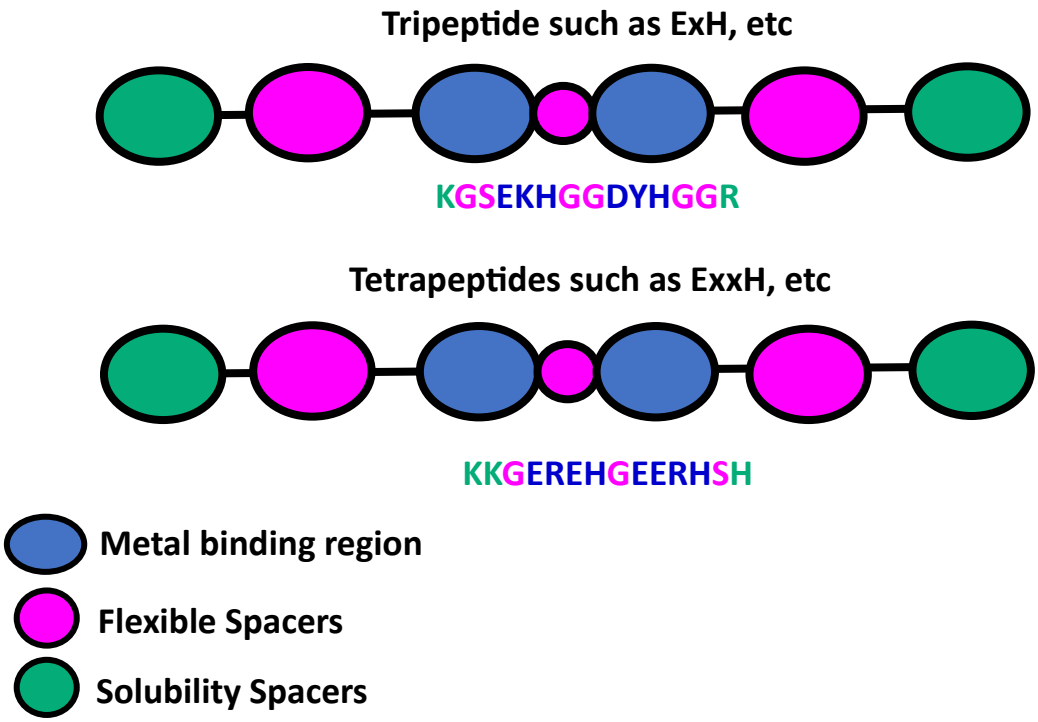


Fig, 7. Peptide template design 1

Crucially, the final peptide construct integrates these binding elements with two distinct types of spacers to optimize physicochemical properties. Solubility Spacers, which are sequences rich in highly charged residues (e.g., poly-Arginine, poly-Lysine, RRREE, KKEEEE, or HHHHH, are used to significantly enhance the peptide's aqueous solubility and stability, preventing aggregation. Simultaneously, Flexible Spacers, composed predominantly of Glycine (G) and Serine (S) (e.g., GGGS, GGGGSGGGGSGGGGS), are incorporated to provide the necessary conformational freedom and distance, ensuring that the acidic and histidine coordinating residues can achieve the precise three-dimensional geometry required for optimal Fe^{3+} chelation.

Template design 2

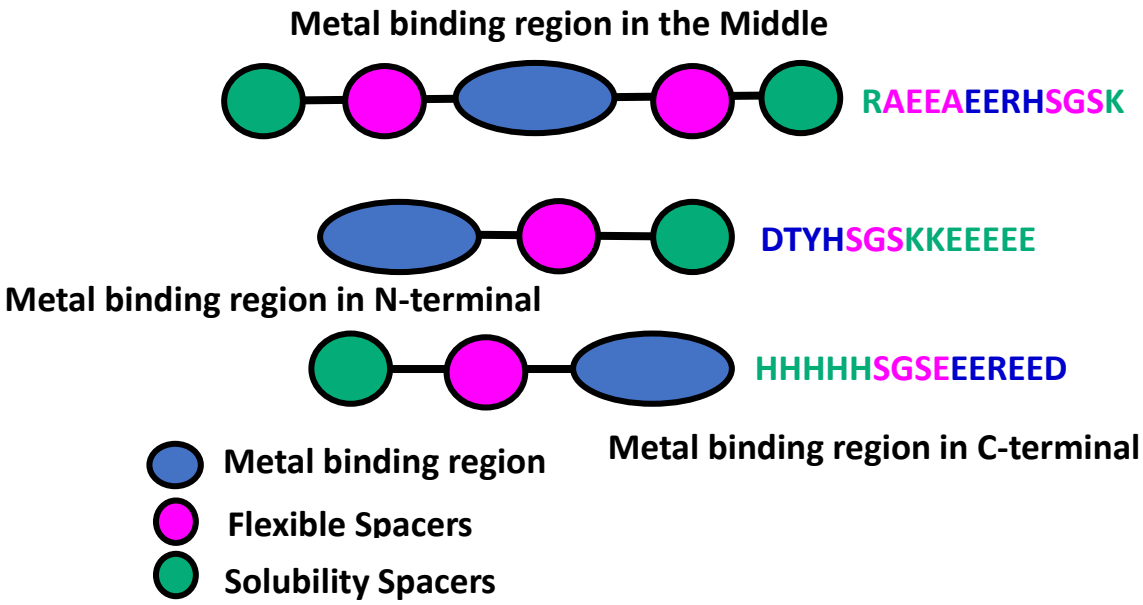
The template shown in Fig 8, systematically organizes the peptide into three functional parts: Metal Binding Regions (Blue Ovals), containing high-affinity Fe^{3+} coordinating residues EKH and ERH which serve as core chelation sites. Amino Acid (AA) Combination Segments (Pink Ovals), which function as Solubility and Flexible Spacers, often rich in highly charged (K, R) or flexible (G, S) residues and Terminal Capping (Green Ovals), representing additional stabilizing or targeting sequences. This hierarchical arrangement frequently creates tandem binding sites (two metal binding regions separated by a pink spacer), ensuring the peptide not only binds Fe^{3+} specifically but also maintains high aqueous solubility and the necessary flexibility for optimal metal-ligand geometry.



Fig, 8. Peptide template design 2

Template design 3

The design shown in Fig 9, incorporates three key modular elements: the Metal Binding Region (blue), which contains the Fe^{3+} coordinating residues (E & H). The Flexible Spacers (pink), composed of residues like Glycine and Serine (e.g., SGS) to ensure conformational freedom; and the Solubility Spacers (green), which are highly charged sequences (e.g., poly-Arginine or poly-Lysine) positioned at the termini to prevent aggregation (e.g., KKEEEEE or HHHHH). This template demonstrates flexibility in positioning the binding core it can be placed in the Middle (e.g., RAEEAERH), the N-terminal (e.g., DTYH), or the C-terminal (e.g., SGEEEREED) region of the peptide. This positional control allows for fine-tuning the peptide's folding, accessibility to the Fe^{3+} ion, and overall function, depending on the final application.



Fig, 9. Peptide template design 3

Clustering

The peptide generation process integrated both reinforcement learning (RL) and a rule-based generation system to produce a diverse set of Fe^{3+} - binding peptides across multiple template designs. To select the most promising candidates, key physicochemical properties including hydrophobicity, pKa, net charge, and molecular weight were calculated. Peptides were then rigorously filtered using machine learning-based clustering and sequence alignment approaches to ensure a high-diversity, high-occurrence set of unique peptide sequences. Recognizing the absence of commercially available Fe^{3+} - chelating peptides for liquid detergent applications, this research employed a broad, multi-strategy approach.

Conclusion

To conclude the project titled ‘Peptide Against Anthocyanins’, aimed at identifying peptides which chelates Fe^{3+} for liquid-detergent applications, we designed a library of thirty (30) peptides and investigated their potential mechanisms of action. Using an integrated workflow combining bioinformatics, molecular modeling, generative AI (reinforcement learning), and rule-based design, we generated diverse peptide templates including tripeptides (3 AA), tetrapeptides (4 AA), and sequences incorporating ferric-ion-binding motifs positioned at the N-terminal, C-terminal, central region, or in hybrid configurations. As no established reference peptides exist for this application, the outcome provides a broad and diverse set of candidate peptides for further evaluation.